ECHO 7 Project Proposal

Autonomous visual docking for MTRX3700

Three-person revision | 22 October 2026

Proposed decision

I propose that we build **ECHO-7: The Last Beacon**, a camera-guided rescue rover that identifies the correct docking station among misleading markers, approaches it and stops accurately. We would use the supplied kit and make our main original contribution a reliable FPGA marker-recognition and tracking pipeline.

This proposal sets out the concept, engineering scope, initial targets, work split and schedule. The team would agree ownership and validate the numerical targets before committing them to the demonstration datasheet. No performance results are claimed yet.

The story

A meteor storm has cut communications with a remote lunar outpost. One rescue capsule is still transmitting, but damaged equipment is producing false beacons. ECHO-7 is the last operational rover. It must identify the real capsule, reach its docking station and establish a rescue link.

In our demonstration, the beacons are printed visual markers and docking means alignment followed by a stop at a defined stand-off distance. The message "Rescue link established" is a simulated mission event. We do not need a physical connector, charging system or gripper.

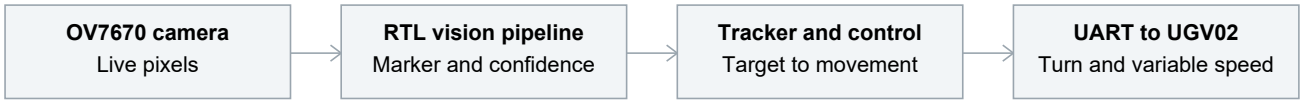
What the audience sees

- The rover starts and searches a bounded area for the authorised marker.
- It rejects similar-coloured decoys and displays the selected target on VGA.
- It approaches and steers towards the docking centreline, reducing speed near the station.
- If an obstacle enters its path, it stops and waits; it resumes only after clear-range and valid-target checks.
- It stops at the docking gap and displays "Docking complete - rescue link established".

Why this design is a strong choice

The task is easy to understand, fits a compact demonstration area and gives us measurable outcomes: docking error, correct-dock completion time and response latency. It can be built in stages, and the same course supports a fair comparison against basic colour tracking. Our aim is a multi-stage original pipeline with evidence; the tutor determines the achievement level actually demonstrated. [1, sections 2.2-2.4]

Technical design and scope



DE1-SoC performs vision, tracking, decisions and command generation.

Ultrasound supplies range; IR stop overrides every motion state.

VGA shows live video, selected marker, confidence, range and mission state.

Laptop records evidence only. Nano remains a communications relay.

Marker and demonstration environment

Start with a flat docking face and a centred marker board approximately 200 mm wide by 140 mm high. A candidate design has a red block on the left and a blue block on the right, each 60 by 80 mm, with 100 mm between their centres. Confirm colours, size and camera visibility during calibration. Decoys use the same colours with swapped order, incompatible spacing or mismatched sizes; identical valid markers are outside the initial scope.

The initial task uses mostly frontal approaches on a flat indoor floor. Place the marker at camera height above a broad face that returns ultrasound reliably. Mark both the bay centreline and a chassis reference point so docking error can be measured independently of the camera.

Processing stages

- **Segment and clean:** compare camera colour channels against calibrated limits, then use a small spatial filter to remove isolated mask noise. Keep per-pixel work in RTL.
- **Extract candidates:** use scanline runs and bounded region descriptors to estimate each colour region's centre, area and extent. Begin with a small candidate capacity, such as eight per colour; flag overflow rather than silently selecting an arbitrary region.
- **Validate and select:** pair regions using colour order, relative size, spacing and vertical alignment. Produce a marker centre, quality value and valid flag.
- **Track and control:** use frame-to-frame consistency and fixed-point smoothing to stabilise the target. Convert horizontal error and valid range into turning and variable-speed commands.

Use the released camera format and UART protocol. A reduced processing resolution, such as 160 by 120 pixels, is an initial option to evaluate rather than an assumed supported mode. Keep clock crossings explicit and agree field widths, reset behaviour, valid strobes and overflow handling before integration. The core can remain entirely in RTL; any later use of the DE1-SoC processor must preserve R4.

Mission states and safety priority

IDLE -> SEARCH -> ACQUIRE -> APPROACH -> FINAL ALIGN -> DOCKED. An obstacle or invalid range causes HOLD. Loss of the target stops translation and returns to a controlled search only when the path and range are valid. IR ABORT overrides every state, clears pending movement and stays latched until a different IR key releases it to IDLE. DOCKED requires a stable valid marker, acceptable alignment, valid gap and zero-speed command; a timeout is a failure, not success.

Optional extension: remember target identity across a brief occlusion and measure reacquisition. Core scope excludes YOLO, off-board inference, arbitrary-object recognition, general maze navigation and driving around obstacles. Add no electronics unless a measured need justifies them and tutors have discussed the addition. [1, sections 2.1-2.2]

Proposed measurable targets

All values below are initial design proposals, not verified performance. We will use early hardware measurements to confirm or revise them, then freeze the test conditions before the final datasheet. Record predicted results from simulation/calculation and measured hardware results separately. [1, section 2.3]

Shared conditions for the docking trials

Stationary authorised marker; start distance 0.7-1.0 m; lateral offset -150 to +150 mm; heading error -10 to +10 degrees; maximum commanded approach speed selected in the 0.10-0.15 m/s range, with lower speed during final approach. Include one or two fully visible decoys, separated from the real marker. Use calibrated camera settings and a declared diffuse indoor lighting setup without glare or direct sunlight. The tutor chooses positions and settings within the declared ranges.

Target	Threshold	Goal
T1 Worst lateral docking error	80 mm or less	40 mm or less
T2 Worst correct-dock completion time	18 s or less	12 s or less
T3 Worst camera-to-command latency	200 ms or less	100 ms or less

T1 Final lateral docking error

Measure the absolute sideways distance between the marked chassis reference point and the bay centreline after stopping. Run three approaches under the shared conditions; the largest error is the result. A wrong bay, collision, or failure to complete within 18 seconds is a failed trial, not an omitted reading. Use floor tape, a ruler and a fixed reference convention. This tests the localisation and steering pipeline together.

T2 Correct docking with decoys

Time from the mission-start command to a stable stop at the correct station. Completion requires lateral error no greater than 80 mm, a front-chassis-to-dock gap of 250-350 mm, and a stopped state for at least 0.5 seconds. Calibrate the ultrasound mounting offset to this physical gap. Run three approaches; the longest time counts, and any wrong station or incomplete trial fails the target. Rehearse three 18-second trials plus resets within approximately 90 seconds. This is our main test of marker validation in the motion loop.

T3 Camera to command latency

In APPROACH with valid range and motion permitted, move the fully visible marker at 0.7-1.0 m between centre positions at 20-35% and 65-80% of image width. Keep a decoy visible; both ranges must be outside the calibrated steering deadband. Measure from the frame-start timestamp of the first frame containing the changed position to the first UART bit of the correctly responding command. Correlate recorded frames, frame IDs and SignalTap/UART capture; do not start timing after detection. Make three changes; the longest delay counts. An absent or wrong-direction response fails.

Fair bypass comparison

A DE1-SoC switch selects a reasonable raw colour-centroid baseline instead of the advanced cleanup, marker-selection and tracking path. Both paths feed the same motion controller and keep camera input, ultrasound, UART and IR safety active. Repeat T2 and, where time allows, T1 at matched positions with identical decoys. Record both results; do not deliberately weaken the baseline or assume it will fail. If it still meets the threshold, the claimed contribution must be reconsidered under R9.

Optional fourth target, only after validation: correct-marker reacquisition after a 0.2-0.4 s occlusion. Three trials; longest time counts. Proposed threshold 0.8 s and goal 0.4 s, with defined stop behaviour during loss.

Verification and assignment evidence

Verification starts alongside implementation. Each module gets self-checking tests before it reaches the robot, and integration tests exercise complete missions and failures. Evidence must explain our own processing contribution, including why simulation and hardware results differ. [1, sections 2.2, 2.5 and 4]

Requirement	Planned evidence
R1 Camera in the loop	Move the live marker and show the corresponding change in steering from DE1-SoC-computed image quantities.
R2 FPGA chassis control	Demonstrate UART direction, turning and variable-speed commands issued by DE1-SoC.
R3 Safety stop	Test IR abort in every state; verify the latch and release key. Demonstrate this before any driving acceptance test.
R4 On-board processing	Show the RTL processing and control architecture. Laptop performs display/logging only; Nano remains a relay.
R5 Measurable targets	Publish 3-5 targets, including timing and accuracy/repeatability, with conditions, three-trial scoring, thresholds and goals.
R6 Automated tests	run_all_tests.sh uses Verilator 5.050, runs module and system tests, prints PASS/FAIL, prints CAUGHT for a deliberately broken module, and finishes within 5 minutes from a clean copy.
R7 Golden model and replay	Compare a self-designed RTL module against a Python model using recorded real-hardware inputs.
R8 Ownership	OWNERS.md maps every source file to an SID and origin. Each member owns substantive synthesizable RTL and a testbench.
R9 Bypass test	Use the physical switch, preserve the running robot and safety path, and repeat the target that measures the bypassed contribution.

Golden model and replay plan

Model the marker validator in Python. Record actual candidate inputs and validator outputs from the FPGA, including frame IDs and valid flags. Replay the captured inputs into both model and RTL, comparing acceptance, selected centre and quality using explicit fixed-point rules. Show the recorded board outputs beside model and replayed-RTL results. Store captures, model, replay driver and comparisons together. These descriptors are real module inputs; synthetic frames remain useful additional tests.

Tests and engineering records

Cover isolated noise, frame boundaries, multiple candidates, invalid geometry, candidate overflow, target loss, invalid range, controller saturation and IR priority. Deliberately break a geometry comparison in an isolated test variant and show the suite catches it; the demonstration bitstream remains the correct version. Capture predicted and actual ALM, M10K and DSP use, Fmax/slack and latency from Quartus and hardware measurements. Keep a dated log of bugs, symptoms, fixes and regression tests, including at least two genuine bugs for the report.

Proposed work split: three people

The work is divided into three connected packages. Member A, B and C are placeholders for the team's SIDs. Each person owns new or substantively adapted synthesizable RTL and its self-checking testbenches; documentation, modelling or wiring alone does not satisfy R8. [1, R8]

Owner	RTL and self-checking tests	Integration and evidence
Member A Camera and candidates	**Own:** colour segmentation, lightweight streaming cleanup, bounded candidate extraction and the raw colour-centroid baseline. **Test:** noise, edges, frame/reset behaviour, split/merge cases, descriptor arithmetic, capacity overflow and throughput.	Camera capture and colour calibration; pixel/candidate clock crossings; descriptor and frame-ID capture taps; early Quartus resource and timing checks. Keep the baseline tap before advanced cleanup.
Member B Recognition and tracking	**Own:** marker geometry validation, candidate selection and temporal tracking/smoothing. **Test:** colour order, size/spacing limits, competing candidates, noisy targets, stale frames and target loss.	Python validator model and real-hardware replay/comparisons; target interface with A/C; VGA target/status overlay integration. Assemble the datasheet/report from everyone's evidence. Independently review C's safety logic.
Member C Motion and safety	**Own:** mission/motion FSM, steering and speed, range-validity gating, latched IR abort and physical-switch path selector. **Test:** state transitions, saturation, loss, invalid/stale range, UART readiness and abort priority.	UART/chassis and ultrasound integration; shared regression runner and system tests; bitstream and code/submission coordination. Preserve the same controller and safety path in baseline and advanced modes.

Interfaces and parallel development

A to B: candidate count, centre/area/extent, frame ID, valid strobe, capacity and overflow. A also defines pixel format, line/frame markers, reset and clock crossings.
B to C: selected/tracked centre, quality, validity and frame ID; A's baseline uses the same target format. C selects either path before the common motion controller.
Control: turn/speed, fresh valid range, motion permission, abort latch and UART readiness. Agree widths, fixed-point rules, stale-data behaviour and resets before integration.

Integration and workload balance

C coordinates a working main-branch baseline. B develops with synthetic candidate descriptors while C develops with target vectors, so neither waits for camera completion. Each owner supplies example input/output data and passing module tests; all three run regression and a short hardware check at each handoff. B reviews and witnesses C's IR stop tests before driving acceptance tests.

A carries the largest pixel-processing workload: keep cleanup modest and extraction bounded, and check resource use early. B leads evidence/report assembly; C leads build/submission packaging. All three write their own report sections, record their video contribution and rotate trial setup, measurements and logging. Cosmetic extras and occlusion recovery wait until the core passes.

Team agreement at handover

Assign the three SIDs and file ownership in OWNERS.md; reconcile pitch feedback; inspect starter code and the Lookbook; book shared hardware sessions; confirm marker geometry, candidate limits, target ranges and the props budget. Discuss any added hardware/connectivity with tutors.

Schedule and demonstration plan

Three-person revision: 22 October 2026. The earlier 16-22 October milestones are a status checkpoint, not a claim of completion. Reconcile the 13 October video pitch and 14 October lecture pitch feedback. Official dates below are carried over from the original proposal; times are Sydney local. [1, sections 3-7]

Date	Three-person milestone or deliverable
22 October Status checkpoint	All: confirm scope, three owners, interfaces and starter-code origins. A: camera/raw-centroid baseline. B: validator/tracker test harness and replay schema. C: supplied hardware test, verified IR stop and basic camera-to-wheel control. Record any unfinished 16-22 October work.
23-26 October	A: cleanup, bounded extraction and baseline target output. B: validator/tracker, capture taps with A, and initial overlay. C: motion integration and physical bypass switch. All: validate both paths and safety; obtain first T1-T3 measurements.
27-30 October	B leads real-input model/replay comparisons and the deliberately broken validator test. A checks timing/resources and candidate captures. C completes system tests and the clean-copy regression runner. All: close interface bugs and freeze core scope by 30 October.
31 October- 2 November	All: repeat three-trial T1-T3 and matched bypass trials; rotate setup, measurement and logging. B assembles the datasheet. A/C verify calibration and motion conditions. Freeze conditions, rehearse and record each person's video contribution.
3 November	**Suggested internal deadline:** C assembles draft Ed code upload, bitstream and package; B checks datasheet/replay evidence; all verify OWNERS.md, review the completed five-minute video and approve their contributions.
4 November 11:59 pm	**Official video deadline:** submit the five-minute recorded presentation. C coordinates submission; all three contribute and check the final file.
5 November	**Official live demonstration and individual Q&A.** A explains the image/candidate pipeline; B the validator/tracker and replay; C control, safety and system integration. Everyone explains their owned code.
6-8 November	All write and review their report sections. B assembles the report and evidence; C packages final code. **Official report/final-code deadline: 8 November, 11:59 pm.** Each member completes SparkPlus and their AI/decision contribution.
9-13 November	All inventory the complete kit and agree who returns it in good condition.

Presentation and live demonstration

Use the five-minute video for the mission, architecture, decisions, ownership, predicted/measured targets and limitations, with all three speaking. At the start of the slot, C launches run_all_tests.sh from a clean copy with the tutor watching while the video plays.

The ten-minute live slot retains about two minutes for tests/model/replay, seven for acceptance and one for a limitation. Start acceptance with IR safety, then the complete mission. The tutor selects three targets and conditions within the datasheet ranges; include the bypass comparison. A handles marker/course setup, B records measurements and presents replay, and C operates the rover and stop control. Rehearse the mission to around one minute.

What we bring and submit

Bring the charged robot, stable marker stands, decoys, tape, ruler, one-page datasheet and evidence laptop. Package Quartus files/constraints, RTL, tests, software, final .sof, run_all_tests.sh, OWNERS.md, golden model/replay data and datasheet PDF. Upload draft code before demo/Q&A; 3 November is an internal target, not a new official cutoff.

The final report is at most six pages, uses SID only, and covers datasheet, design, decisions, verification, hardware budget and debugging. The contributions table and one-page AI/decision Statement are outside that limit. Everyone prepares for individual Q&A.

Resources risks and decisions

Hardware and presentation resources

Use the supplied DE1-SoC, UGV02, Robot Adaptor Board, OV7670 camera, RCWL-1601 ultrasound, IR remote and VGA setup. Keep the Nano in its relay role. No additional sensor purchase is planned. Build the lunar capsule from recycled cardboard or printed panels, add a removable ECHO-7 label and keep the marker clearly lit. The proposed A$40 cap covers props and printing; the team must agree it.

Main risks and fallbacks

Risk	Response and lead
Camera colour changes or false marker matches	**A/B:** calibrate under declared lighting, use geometry as well as colour, and test recorded difficult scenes. Keep marker definitions precise.
Image-centred target but inaccurate physical docking	**A/C:** fix the camera mount, calibrate camera/chassis offsets, use a physical centreline and restrict initial approach angles.
Ultrasound measures a prop or misses an angled face	**C, reviewed by B:** use a broad flat docking face, calibrate mounting offset and stop on invalid/stale range. Test props before the demo.
Candidate extraction or timing exceeds the hardware budget	**A:** bound candidates, keep cleanup small, evaluate supported processing resolution and check Quartus early. Preserve the tested baseline.
Three-person workload or integration slips	**All; C coordinates:** review workload at every handoff and help with fixtures, captures and regression. By 30 October, drop optional occlusion recovery and cosmetic extras first; preserve core safety, RTL ownership and evidence.
Bypass still meets the claimed target threshold	**B with A/C:** investigate whether processing solves a real problem. Improve the design or honestly revise the contribution/claim; never rig the baseline.

Equipment and academic responsibilities

Run the supplied hardware test and report faults. Use supplied rails and removable mounts; leave the protective Plexiglas intact. The supplied ultrasound is 3.3 V only: check polarity and follow the adaptor instructions, including never wiring J5's 3V3 pin to the chassis. Discuss extra components, connectivity or hazards with tutors before use. [1, sections 2.1 and 9]

Attribute staff, earlier-assignment and external code in OWNERS.md; do not use another group's code. Each owner must understand their implementation. Retain relevant AI conversation exports for the report and prepare the required one-page statement describing tools, representative prompts, an error or limitation and verification. AI assistance is not allowed during the supervised demonstration or individual Q&A. [1, sections 2.5 and 8]

Proposal for the three-person team to agree

Adopt ECHO-7 as a precise visual-docking mission with marker validation under distractions as the central contribution. Allocate A to camera/candidates, B to recognition/tracking and C to motion/safety. Use the 22 October checkpoint to establish the actual baseline status and next handoff. Keep the three core targets and all verification evidence; add occlusion recovery only after the core passes.

Source

[1] MTRX3700 2026 Major Project Brief, cited in the original proposal as MTRX3700_2026_Major_Project_Brief_final-1.pdf. Requirements/targets: pages 3-4; achievement/implementation: pages 4-5; demonstration/submissions: pages 6-7; hardware: pages 1-2. Requirements and official dates are carried over; the brief was not supplied for this revision. Story, architecture, targets, allocations, budget and internal milestones remain proposals for team validation.